

LIFESTYLE FACTORS

Although lifestyle factors are not the primary focus of this book, they are very important to your training.

SLEEP QUALITY

Skimping on sleep is a surefire way to decrease progress with both strength and hypertrophy. Sleep is vital for recovery. While you sleep, your body prepares for your next workout. Sleep is the most anabolic time during the day.

The amount of sleep needed each night will vary from one person to another, ranging from six to nine hours. Most people do well with seven and a half to nine hours. Sleep cycles are typically an hour and a half in length, so it is actually more beneficial to get seven and a half or nine hours of sleep than eight hours. Your sleep cycles may be a bit longer or shorter: spend some time evaluating what works best for you. It is important, because waking up mid-cycle will cause you to feel groggy. It is important to wake in a lighter sleep rather than in the middle of a deep sleep. You can easily track your sleep cycles through free or paid iPhone and Android apps; there are even alarm clock apps that will wake you during your lightest period of sleep within a set window. Beyond this, there are numerous ways you can improve your sleep. Here are a few quick tips:

- Use *Flux* or a similar dimming application for your computer/phone that emits red light after dark in order to not mess with circadian rhythms.
- Limit use of electronic devices one to two hours before bedtime. Use this time to read or meditate.
- Eliminate electronic devices, outlets, and plugs near your bed.
- Sleep in a pitch-black room with blackout curtains, or use a sleep blindfold.
- Eliminate all sounds; use earplugs if needed. Alternatively, some people sleep better with white noise (rain-type sounds or fans) than utter silence.
- Sleep in a cool, dry room (about 60-65 degrees if possible).
- The best scenario is getting to bed early enough that you wake without an alarm. Try to go to bed before 10:00.
- Keep your sleep habits consistent. It is better to go to bed consistently at a later time than jumping around (10:00 one night, 1:00am the next, 11:00 the night after).
- Avoid caffeine and other stimulants before bed.
- Avoid alcohol and other depressants before bed. Although alcohol can help you fall asleep faster, your sleep quality will usually suffer.

Here are some physical exercises and other techniques that can help improve sleep quality:

- Stand on one leg until you are exhausted, then switch to the other leg and do the same. This fatigues your hip muscles, which can improve sleep quality.
- Perform Esther Gokhale's spine lengthening routine before bed. Move from sitting to stretching your spine out as you lie down.
- Exhaust yourself from physical activity like hiking, pickup sports, lifting, running, etc.
- Deep breathing exercises.
- Deep tissue massage.
- Light stretching or foam rolling.
- Long, hot showers. (Cold showers are more effective for some people.)
- Plan any exercise, especially high-intensity exercise, earlier in the day rather than later.

These are just a few of the ways you can improve your sleep quality. Experiment with these and others until you find what works best for you. Supplements can be used, but try the natural methods listed above first. If you are already doing many of these things, the following list of supplements can improve sleep but they should be just that—supplements. Always consult a medical professional before using these or any other supplements.

- Medium chain triglycerides from sources like coconut milk.
- Magnesium. Orally works, but it absorbs better topically—via Epsom salt or other magnesium salt baths, especially.
- Melatonin.
- Phosphatidylserine, an anti-cortisol supplement.
- 5-hydroxytryptophan or 5-HTP, a derivative of tryptophan.
- L-theonine.
- Valerian root.

Sleep can be a troublesome issue. Here are two final tips. One is more of a psychological conditioning tip—use the bedroom *only* for sleep or sex. It can be difficult to fall asleep if you hang out in your room a lot and are conditioned to watch TV, play adrenaline-infused games, or study in an alert state there.

The second tip is a personal technique that I utilize. First, settle into a comfortable reclined position. Then, initiate a deep breathing cycle (in through the nose out through the mouth) with a 6/3 cadence—six seconds of inhalation through your nose and three seconds of exhalation through your mouth. Now roll your eyes back in your head as though you are already knocked out from sleeping. Finally, begin actively daydreaming or imagining something cool—something you wish would happen that is not directly related to daily life. This prevents your mind from wandering and getting frustrated and may help transition you into the sleep you desire.

Andrew Weil has a similar technique called the *4-7-8 breathing technique*. To perform this, breathe in through your nose for four seconds. Now, hold your breath for seven seconds. After that, exhale through your mouth for 8 seconds. Repeat the cycle until you are relaxed and fall asleep. Elongating your breathing, paired with holding your breath, helps calm down your nervous system and slow your heart rate for relaxation.

NUTRITION

Since this is not a nutrition book, the only recommendation is that you eat lots of nutrient-rich foods and avoid foods to which you have allergies or sensitivities. This approach works for most people and successfully avoids the polarizing nutrition recommendations of so-called fad diets.

Nutrient-rich foods are prevalent in most cultures: fruits, vegetables, nuts, seeds, meat, birds, fish, eggs, dairy, and grains. High-quality foods contain all of the vitamins and nutrients you need, and also have a relatively low caloric density to help you stay lean. Carbohydrate staples are generally solid as long as you are not allergic or sensitive to them: potatoes, sweet potatoes, rice, yams, and various grains (ideally prepared traditionally through soaking). Animal meats, especially organ meats, tend to be the best way to get protein. You can get good fats from the remaining categories, such as nuts, fish, and dairy.

If you find yourself allergic or sensitive to any particular foods or categories of food, avoid these while attempting to gain as many nutrients as possible from what is left. Consulting a nutritionist may be required, depending on your degree of sensitivity or reaction. There are allergy tests that are made to detect these sensitivities and reactions.

Studies have shown that many athletes benefit from increasing their daily protein intake to .7 grams per pound or about 1.5 grams per kilogram of bodyweight. If you are training for strength or hypertrophy, this or a slightly higher protein intake is recommended. Experiment and find out what works best for you.

Here are four general statements that sum up the biggest concepts that most athletes need to learn. If you have personal weight loss or weight gain (for muscle mass) goals, you may need to make some modifications.

- **Diet modulates weight.** Nutrition is what governs your weight. If you consume more calories than your body actually uses, you will gain weight. In contrast, if you consume fewer calories than your body uses, you will lose weight. To burn fat, cut your calories consumption.
- **Exercise modulates body composition.** Exercise is what causes the body to gain muscle mass and/or use fat for energy. This is especially true on a hypocaloric diet, where you eat fewer calories than you burn in a day. When athletes are cutting to make a weight class, they still work out and eat enough protein so that they will maintain as much muscle as possible. Obviously, diet modulates body composition to some extent too. An obese person losing 50-100 lbs of fat will have improved body composition.
- **Nutrition quality is related to health and partly to weight.** Nutrition quality is specifically related to health in that the more nutrient-dense foods you eat, the healthier you will be. High or low nutritional quality only partly affects people's weight as regards the speed of weight loss on a caloric deficit, unless there are already dysfunctions in metabolism (like thyroid problems).
- **Exercise intensity improves speed of body composition changes.** High-intensity exercises like sprinting, strength and hypertrophy training through barbells or bodyweight, circuit training, etc. tend to improve the speed of body composition changes over lower-intensity exercise like cardio, cycling, and other endurance sports. The fastest way to drop weight is a combination of high-intensity and low-intensity exercise.

Some colloquial statements like “*Abs are made in the kitchen*” are true (although exercise helps). Others like “*high repetitions and low weights for toning*” are false (toning is simply increasing muscle and decreasing fat mass). The fastest way to tone would be to get your diet under control, lose weight, and perform hypertrophy-specific training to increase muscle mass.

- **Calories In = What You Eat**
- **Calories Out = Total Daily Energy Expenditure (TDEE) = Basal Metabolic Rate (BMR) + Activity**

Calories in is a simple concept as it takes stock of what you eat. *Calories out* is only slightly more complicated. The concept of *total daily energy expenditure* (TDEE) is a combination of *basal metabolic rate* (BMR) and calories expended through activity. BMR is a physiological term for the energy it takes to keep you alive while in a state of rest. Most of your energy is expended to keep your body warm and your organs running. Activity accounts for the rest of the calories you burn. With these two concepts in mind, we can begin to talk about weight loss and weight gain.

Weight Loss

If you are obese and change your diet to consist of low calorie, nutrient-dense foods (such as vegetables), you will naturally lose weight. The problem with the modern *standard American diet* (SAD) is it contains very few nutrients per calorie. Potato chips and French fries are low in nutrients, high in calories, and are not very filling. Most sugary foods are like this too. Modern processed grains like spaghetti and bread are similar, though several decades ago they were not. Cut down on these common American foods and consume more fruits and vegetables and traditionally prepared (soaked, slow cooked) grains in their place.

Given that diet modulates weight, you need to consume fewer calories than you are burning. Eating more filling foods will help. Eat until you are satisfied (70-80% full) rather than 100% full. In conjunction with this, eliminate as many empty calories as you can, especially liquids. Replace juices, sodas, and lattes with water, tea, or black coffee.

Tracking calories solves many weight loss issues, as most people routinely underestimate how many calories they actually consume each day, especially through liquids that contain 150-180 calories per serving. In terms of tracking *calories in < calories out* it is best to take two actions during the course of a week. Chart your food intake for the week using a physical or online food journal like FitDay. Divide by the number of days to find out how many calories you typically burn each week. You want to consume approximately 200-500 fewer calories in subsequent weeks if you are trying to lose weight. Alternatively, you can use the Internet to find a TDEE equation that will allow you to estimate your BMR and calories burned from activities. This should give you a ballpark number to start from; decrease calories from there.

Weight Gain

If you are looking to gain weight, the process is super simple: Eat like a madman. In order to gain weight in the context of muscle mass, three things need to be present:

- **High-Intensity Stimulus:** lifting weights, bodyweight strength training, sprinting, and the like. This is necessary to force your body to adapt to add muscle mass.
- **Caloric Excess:** This gives your body the energy needed to produce muscle mass.

- **Enough Protein:** Generally, aim for .7-1 g/lbs protein per day of the weight that you want to be. If you are 150 lbs and want to weigh 200 lbs, aim for 200 g/lbs protein per day.

The only difference between this and what you should do for weight loss is that you increase the amount of calories you ingest instead of decreasing it.

Let's talk about the scenario of losing fat and gaining muscle simultaneously. The general rule of thumb is that extra fat within the body is an energy source that can be metabolized by the body. Therefore, if you are overweight and lifting weights, but eating in a caloric deficit, your body can call on the fat mass as the extra energy needed to build muscle. As your body composition improves you will no longer be able to do this as effectively. However, this is one of the extraordinary capabilities of the body and a reason why lifting weights can be superior for losing fat than cardio or higher repetitions. The higher your percentage of body fat, the easier it will be for your body to lose fat and gain muscle simultaneously.

Workout Nutrition

Workout nutrition is a specific case. Eating one to two hours before working out could either have a positive or negative effect on your workout. Food in the gut may cause parasympathetic nervous system activation, which may interfere with the optimal sympathetic nervous system output needed for exercise. However, it really depends on the individual. Many athletes prefer fasting before a workout, but you should try both methods and do what works best.

A recent meta-analysis by Brad Shoenfeld entitled *The effect of protein timing on muscle strength and hypertrophy* suggests that the timing of protein intake may not matter as much as your total daily protein intake. Drinking a protein shake before, around the time of a workout, or after helps people accumulate the amount of total protein they need for the day. Protein synthesis occurs in approximately 3 hour spikes in the body, so time will tell if more research affirms this. Classically, if you want to optimize muscle mass gain, divide your protein intake into 30-gram portions taken every two to three hours.

Supplementing with liquids instead of eating after a workout does make a difference. Liquid nutrition reaches the muscles faster. While it is preferable, it is not absolutely necessary. Whole chocolate milk is one of the best workout drinks because it contains natural whey, casein, and carbohydrates for energy.

Supplements should be just that—supplements to a good diet. The primary supplements that have been proven to work effectively are whey, creatine, casein, and BCAAs. But use good judgment: research has consistently shown that multivitamins and minerals tend not to be as effective as real food, so eat as much nutrient-rich food as possible. Ensure particularly that you get enough vitamin D from sunlight and fish oil. If you do have a deficiency in any area, supplements can definitely help, though be careful if you self-diagnose: there is not enough data available on the long-term effects of vitamin and mineral supplementation for those who do not have a deficiency. For the latest unbiased research on supplements, visit www.examine.com.

WORKING OUT WHILE SICK

Controversy exists over whether or not you should work out while you are sick. High-intensity workouts place greater amounts of stress on the body, which can exacerbate an illness. The same immune system that helps restore physiological damage from workouts also fights infections and pathogens. If your body is working

overtime to recover from a high-intensity workout, it may have a hard time recovering from an illness. People who are sick often perform a workout and become sicker, and people who are nearly recovered from an illness might perform a workout, only to have their illness return a few days later.

It is not generally a good idea to perform resistance training, high-intensity interval training, circuit training, or sprinting when you are sick or recovering from an illness. Allow your body to fully focus on recovery, rather than diverting your immune system to repair muscles post-workout. If you do decide to work out, some low-intensity resistance or strength training can be fine, assuming that the damage done to your muscles is minimal. High-repetition endurance training can also be acceptable as long as the overall volume is not too high. The type of training that does the most muscle damage is moderate to heavy weight and moderate repetitions in the six to twelve range; this is what should be avoided.

It is widely held that it is okay to work out if you have only a head cold, but you should not work out if the illness is in your throat, chest, or lower body. This holds true, as long as you keep the intensity of the workout low. However, if you have a fever, you should not work out at all. Allow your body to use all of its resources to fight off infection while you are in this vulnerable state.

Chest illnesses like pneumonia and bronchitis are more potent than head colds. Doctors prefer that you stay in bed while recovering from these illnesses. However, this can create an issue for an athlete. Total rest and immobility (i.e., bed rest) can foster these conditions further. While you are in a hospital on bed rest, you are at risk of nosocomial infections. At such a time, your body needs some type of movement aside from an actual workout, which it could not handle. You may want to get up to take a short walk to get your blood flowing, or perform deep breathing exercises, especially if you are coughing a lot. This is better than lying there, doing nothing.

If you feel like your body is up for a workout, keep the intensity light. Perhaps you could go for an easy recovery run, or perform a warm-up routine. Pay close attention to how you feel and take notes in your workout book. If you have the tendency to overdo it once you get warmed up, avoid workouts completely until you are back to full health.

Any exercise you perform should leave you feeling better than you started. If you begin to feel worse, stop exercising and rest for the remainder of the day. Remember, your body and muscles recover when you rest, and the same is true for recovering from illnesses.

CHAPTER 16 SUMMARY

LIFESTYLE FACTORS

Sleep quality and nutrition have a big influence on the body, so you should do everything you can to ensure you get enough sleep and sufficient nutrients from your food.

In general, diet modulates weight. Exercise modulates body composition. **Nutrition quality is related to health and partly to weight. Exercise intensity improves speed of body composition changes.**

Weight gain and loss is a product of calories in and out. For athletes, aim for .7-1 g/lbs per day of protein. Modify your caloric intake to your needs in terms of weight loss or weight (muscle) gain.

Supplements are not recommended as a rule, as it is more beneficial to get the majority of your nutrients directly from actual food. However, supplements can help if you develop a deficiency in a certain area. It is particularly important to ensure your body gets enough vitamin D through sunlight and omega 3s such as in fish oil. If you are sick, avoid working out; if you do choose to work out, keep it light.